

ROYAUME DU MAROC • ENSEIGNEMENT SECONDAIRE QUALIFIANT

Pedagogical Lesson Plans

Physics & Chemistry (2nd Year Baccalaureate - Sciences)

Experimental Sciences (Physics-Chemistry, SVT, Agronomy) &
Mathematical Sciences (A & B)

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Registration No / N° SOM : 1503811

Approved Textbooks – Pedagogical Orientations – Ministerial Notes 09-142 & 144

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Introduction & Part 1: Waves (Total: 15 - 16 Hours)

Questions Posed to the Physicist (1h): Identify role of physics in society, scientific questions, measurement uncertainties, and scientific models.

Part 1 Competencies & Units: Wave models for mechanical and light waves. Ultrasonic waves applications. Simulation tools. 1. Progressive Mechanical Waves (5h) • 2. Periodic Mechanical Waves (5h) • 3. Propagation of a Light Wave (5-6h)

Unit 1: Progressive Mechanical Waves			5 Hours
Prerequisites: Relation $v = d / t$, oscilloscope time base, mechanical energy.		Problem Situation: A disturbance propagates through an elastic medium causing temporary local alterations. What defines mechanical waves and their time delay?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Qualitative evidence of 1D, 2D, and 3D waves. - Transverse waves (rope, water) vs longitudinal waves (sound, spring). - General wave properties (propagation, superposition, non-transport of matter). - Propagation velocity $v = d / \Delta t$ and affecting factors (tension, density). - Time delay $\tau = d / v$ and motion relation $y_M(t) = y_S(t - \tau)$.	I. Wave Examples (1. Transverse wave, 2. Longitudinal wave) II. Properties of Progressive Waves (1. Definition, 2. Propagation direction, 3. Velocity expression) III. 1D Progressive Wave & Time Delay (1. Time delay τ , 2. Elongation relationship)	- Textbook, board, computer animations - Rope, spring, ripple tank, tuning fork - Bell jar with vacuum pump, 2 microphones, oscilloscope	Activities: Demonstrate sound non-propagation in vacuum. Record signals from 2 microphones to compute sound velocity in air ($v \approx 340$ m/s). Evaluation: Diagnostic tests, time delay calculations, Exam 1.

Professor: **AYOUB KHAMMOUR** (SOM: 1503811)

References: Approved Textbooks – Pedagogical Orientations – Ministerial Memos 09-142 & 144

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Unit 2: Periodic Mechanical Waves		5 Hours	
Prerequisites: Progressive mechanical wave, velocity, time delay, oscilloscope.		Problem Situation: Ocean swells act as sinusoidal waves. What characterizes double periodicity, and what happens when waves encounter a slit comparable to wavelength?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Sinusoidal wave: double periodicity T (temporal) and λ (spatial). - Fundamental wave equation: $\lambda = v \cdot T = v / N$. - Points oscillating in phase ($d = k \cdot \lambda$) and in phase opposition. - Mechanical diffraction condition: slit aperture $a \leq \lambda$. - Dispersive medium: velocity depends on frequency N.	I. Periodic Wave (1. Definition, 2. Velocity, 3. Stroboscopic observation) II. Sinusoidal Wave (1. Wave along string, 2. Water ripples (circular & plane), 3. Sound waves) III. Diffraction Phenomenon (1. Condition $a \leq \lambda$, 2. Properties of diffracted wave) IV. Dispersive Media	- Ripple tank with mechanical vibrator, stroboscope - Adjustable barrier slit diaphragms - GBF generator, loudspeaker, 2 microphones, oscilloscope	Activities: Freeze water ripples with stroboscope to measure λ . Demonstrate diffraction through slit. Vary frequency to prove water is dispersive. Evaluation: Phase matching exercises on oscilloscope and Exam 1.

Unit 3: Propagation of a Light Wave		6 Hours	
Prerequisites: Mechanical wave diffraction, refractive index, Snell-Descartes laws.		Problem Situation: Laser beams measure microscopic dimensions of hair or wire fibers with micron precision. How does diffraction establish the wave nature of light?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Wave model of light proven by diffraction. - Diffraction angular half-width: $\theta = \lambda / a$. - Relation with central fringe width: $L = 2 \cdot \lambda \cdot D / a$. - Monochromatic vs polychromatic light; white light spectrum. - Frequency ν is invariant across media; speed $v = c / n$. - Dispersion by prism and Snell-Descartes refraction.	I. Diffraction of Light (1. Wave model, 2. Slit diffraction, 3. Formula $\theta = \lambda/a = L/(2D)$) II. Dispersion of Light (1. Refractive index $n(\nu)$, 2. Prism dispersion, 3. Prism formulas)	- Red He-Ne or diode laser (650 nm), green laser - Calibrated slits ($a = 0.04$ to 0.2 mm), thin wires, screen - Glass prism, goniometer, white light source	Activities: Project laser diffraction pattern; measure central fringe width L vs distance D and slit width a ; deduce laser wavelength λ . Evaluation: Linear regression $\theta = f(1/a)$ problems and Exam 1.

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Part 2: Nuclear Transformations (Total: 10 - 14 Hours)

Unit 1: Radioactive Decay		5 Hours	
Prerequisites: Atom structure, isotopes, atomic number Z, neutral atom.		Problem Situation: Carbon-14 dating determines the age of ancient fossils and archeological relics. What causes radioactive decay and what laws govern it?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Nuclide notation ${}^A_Z X$ and Segrè chart (N vs Z stability valley). - Soddy's conservation laws for nucleon number A and charge Z. - Types of spontaneous decay: α, β^-, β^+, and γ de-excitation. - Radioactive decay law: $N(t) = N_0 \cdot e^{(-\lambda t)}$. - Half-life: $t_{1/2} = \ln(2) / \lambda = \tau \cdot \ln(2)$ and activity $A(t) = \lambda \cdot N(t)$ (Bq). - Carbon-14 archaeological dating calculations.	I. Nuclear Stability (1. Nuclide symbol, 2. Isotopes, 3. Segrè diagram) II. Radioactivity Types (1. Soddy's laws, 2. α , β^- , β^+ decays, 3. γ radiation) III. Decay Law (1. $N(t) = N_0 e^{(-\lambda t)}$, 2. Half-life $t_{1/2}$, 3. Activity $A(t)$) IV. Dating (1. Carbon-14, 2. Geological dating)	- Geiger counter and background radiation sensor - Segrè chart poster / computer chart - Dice model simulation program for decay law	Activities: Simulate radioactive decay with dice rolling; plot $N(t)$ curve; deduce half-life. Balance decay reactions using Soddy's laws. Evaluation: Decay equations and dating calculation exercises.

Unit 2: Nuclei, Mass and Energy		5 - 9 Hours	
Prerequisites: Radioactivity, conservation laws, energy units.		Problem Situation: Nuclear reactors release immense energy from Uranium-235 fission, while stars fuse hydrogen. What is binding energy and the Aston curve?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Mass-energy equivalence: $E = m \cdot c^2$. - Mass defect: $\Delta m = [Z \cdot m_p + (A - Z) \cdot m_n] - m_{\text{noyau}} > 0$. - Nuclear binding energy: $E_I = \Delta m \cdot c^2$ (MeV) & E_I / A. - Aston curve analysis ($-E_I/A$ vs A); nuclear fission and fusion. - Energy balance of nuclear reaction: $\Delta E = [\sum m_{\text{prod}} - \sum m_{\text{reac}}] \cdot c^2$.	I. Mass-Energy Equivalence (1. Mass defect Δm , 2. Binding energy E_I , 3. Binding energy per nucleon, 4. Aston curve) II. Energy Balance of Nuclear Reactions (1. General reaction, 2. Spontaneous decay, 3. Fission and fusion) III. Nuclear Energy Hazards & Uses	- Official Aston curve chart, mass tables in u and MeV/c^2 - Video documentary on nuclear power plants & tokamak fusion - Scientific calculators for mass defect computation	Activities: Compute binding energy of Helium-4 and Iron-56. Calculate released energy $E_{\text{lib}} = \Delta E $ for U-235 fission and D-T fusion. Evaluation: Nuclear reaction energy balance problems and Supervised Exam 2.

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Part 3: Electricity (Total: 22 - 38 Hours) • Unit 1: The RC Dipole

Competencies: Understand transient electrical regimes, differential equations, energy storage in capacitors and inductors, and electrical resonance.

Part Units: 1. RC Dipole (6-7h) • 2. RL Dipole (6-7h) • 3. Free RLC Oscillations (8h) • 4. Forced Oscillations (8h - SM) • 5. EM Waves & Communication (10h - SM)

Unit 1: The RC Dipole

7 Hours

Prerequisites:

Ohm's law, node/mesh laws, ammeters, voltmeters, oscilloscope.

Problem Situation:

Camera flash units discharge a capacitor through a xenon flash lamp. What is a capacitor, how does it charge, and what is its time constant?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Capacitor definition, symbol, and charge $q = C \cdot u_C$ (Farads F). - Current-charge relation: $i(t) = dq/dt = C \cdot du_C/dt$. - Equivalent capacitance: parallel $C_{eq} = \sum C_i$; series $1/C_{eq} = \sum 1/C_i$. - Establish and solve RC differential equation: $u_C + RC \cdot (du_C/dt) = E$. - Time constant $\tau = R \cdot C$ (seconds); graphical determination (tangent, 63%). - Electrostatic energy stored: $E_e = 0.5 \cdot C \cdot u_C^2$ (Joules).	I. The Capacitor (1. Description, 2. Charge relation $q = C \cdot u_C$, 3. Association in series & parallel) II. RC Response to Voltage Step (1. Charging differential equation, 2. Analytical solution $u_C(t) = E(1 - e^{-(t/\tau)})$, 3. Discharge differential equation, 4. Time constant $\tau = RC$) III. Stored Energy (1. Formula $E_e = 0.5 C u_C^2$, 2. Flash lamp discharge)	- Capacitor decade boxes (1 μF to 100 μF), variable resistors - DC power supply (6V, 12V), square-wave generator (GBF) - Digital storage oscilloscope / PC data acquisition interface	Activities: Display charge/discharge curves on oscilloscope; determine τ using origin tangent; verify dimensional analysis of $[\tau] = [T]$. Evaluation: Differential equation solutions and energy calculations; Exam 2.

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Unit 2: The RL Dipole		6 Hours	
Prerequisites: Ohm's law, RC circuit, magnetic field, oscilloscope.		Problem Situation: Coils oppose sudden current changes, producing high inductive voltages. What is a coil, what is inductance L, and how does current establish?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Coil voltage equation: $u_L(t) = L \cdot (di/dt) + r \cdot i$. - Inductance L in Henries (H); internal resistance r (Ω). - Differential equation for current setup: $L \cdot (di/dt) + (R+r) \cdot i = E$. - Time constant: $\tau = L / (R + r)$ (seconds). - Magnetic energy stored: $E_m = 0.5 \cdot L \cdot i^2$ (Joules).	I. The Inductor Coil (1. Description, 2. Voltage $u_L = L \cdot di/dt + r \cdot i$, 3. Measurement of L and r) II. RL Transient Response (1. Current establishment, 2. Differential equation, 3. Time constant $\tau = L/R_{tot}$, 4. Current break & diode freewheeling) III. Magnetic Energy (1. Formula $E_m = 0.5 L i^2$)	- Inductor coils with iron core, resistors - Diode (for freewheeling protection) - Square-wave generator GBF, oscilloscope, computer interface	Activities: Observe delayed current establishment on oscilloscope; determine τ ; measure coil inductance L and internal resistance r. Evaluation: RL differential equations and dimensional analysis tests.

Unit 3: Free Oscillations in a Series RLC Circuit		8 Hours	
Prerequisites: RC and RL dipoles, differential equations, energy.		Problem Situation: Radio transmitters and receivers contain electrical oscillators. How do RLC circuits oscillate, why do they damp out, and how are oscillations sustained?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Three oscillation regimes: pseudoperiodic, aperiodic, critical. - Ideal LC circuit differential equation: $d^2u_C/dt^2 + (1/LC) \cdot u_C = 0$. - Natural period: $T_0 = 2\pi \cdot \sqrt{L \cdot C}$ (seconds). - Total energy exchange: $E_T = E_e + E_m$ and Joule dissipation. - Sustaining oscillations using an op-amp negative resistance converter.	I. Capacitor Discharge through Coil (1. RLC circuit setup, 2. Three damping regimes) II. Analytical Study of Undamped LC Circuit (1. Differential equation, 2. Solution $u_C(t) = U_m \cdot \cos(2\pi t/T_0 + \phi)$, 3. Period $T_0 = 2\pi\sqrt{LC}$) III. Energy Exchanges & Maintenance (1. Energy transfer curves, 2. Active maintenance)	- Inductor coils, capacitor decade box, resistance box - Operational amplifier (741) maintenance circuit with $\pm 15V$ supply - Digital oscilloscope, interface and software (Regressi)	Activities: Display pseudoperiodic oscillations; verify $T \approx T_0 = 2\pi\sqrt{LC}$. Connect negative resistance circuit to observe sustained undamped sine wave. Evaluation: RLC energy dissipation calculations and Supervised Exam 3.

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Part 3: Electricity • Units 4 & 5 (Mathematical Sciences Stream)

Unit 4: Forced Oscillations in a Series RLC Circuit (SM) 8 Hours

Prerequisites: Series RLC circuit, AC sinusoidal signals, phase difference.		Problem Situation: Tuning a radio selects one exact frequency among thousands. What is electrical resonance and impedance in forced RLC circuits?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Exciter (GBF generator) and resonator (RLC circuit). - Circuit electrical impedance: $Z = U_m / I_m = \sqrt{R_{tot}^2 + (L\omega - 1/C\omega)^2}$. - Electrical current resonance: $\omega_0 = 1/\sqrt{LC} \Rightarrow N_0 = 1/(2\pi\sqrt{LC})$. - Resonance bandwidth at -3dB: $\Delta N = N_2 - N_1 = R_{tot} / (2\pi \cdot L)$. - Quality factor: $Q = N_0 / \Delta N = L \cdot \omega_0 / R_{tot}$.	I. Forced Oscillations (1. Exciter & resonator, 2. Impedance Z, 3. Phase shift φ) II. Current Resonance (1. Resonance condition $L\omega = 1/C\omega$, 2. Resonance curve $I(N)$, 3. Bandwidth ΔN & Quality Q)	- Variable frequency sine generator (GBF 10 Hz - 100 kHz) - Inductor coil, capacitor, calibrated resistors - Dual-trace oscilloscope and digital AC multimeter	Activities: Plot current I_m vs frequency N ; find resonance peak N_0 ; measure bandwidth at $I_m/\sqrt{2}$ and compute quality factor Q. Evaluation: Impedance and resonance bandwidth calculations.

Unit 5: Production of Electromagnetic Waves & Communication (SM) 10 Hours

Prerequisites: AC voltages, resonance, RLC circuits, multiplier circuit.		Problem Situation: Radio antennas transmit information across thousands of kilometers. Why is high-frequency carrier modulation required, and how is AM demodulated?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Principle of Amplitude Modulation (AM): carrier $p(t)$ and signal $s(t)$. - Modulated voltage: $u(t) = k \cdot [s(t) + U_0] \cdot p(t) = A \cdot [1 + m \cdot \cos(2\pi f_s t)] \cdot \cos(2\pi F_p t)$. - Modulation rate: $m = (U_{max} - U_{min}) / (U_{max} + U_{min}) < 1$ for good modulation. - Trapezoidal method for checking modulation quality. - Demodulation: diode envelope detector (R-C condition: $T_p \ll \tau \ll T_s$).	I. Need for Modulation (1. Antenna dimension problem, 2. Frequency multiplexing) II. Amplitude Modulation (AM) (1. Multiplier circuit, 2. Modulation rate m, 3. Trapezoidal check in XY mode) III. Demodulation (1. Envelope detection, 2. Elimination of DC component)	- Analog multiplier IC (AD633), two GBF generators - Signal diode (1N4148), RC filter components, loudspeaker - Dual-channel oscilloscope with XY display mode	Activities: Modulate low-frequency sound on 20 kHz carrier; display trapezoid in XY mode; tune envelope detector RC filter to recover audio. Evaluation: Modulation rate calculations and filter condition problems.

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Part 4: Mechanics (Total: 35 - 47 Hours) • Units 1 & 2: Newton's Laws & Vertical Fall

Competencies: Apply Newton's laws to predict kinematics and dynamics of solids. Prevent traffic accidents. Use numerical integration (Euler's method). Model everyday mechanics.

Part Units: 1. Newton's Laws (5-6h) • 2. Vertical Fall (3-5h) • 3. Planar Motions (4-5h) • 4. Satellites & Planets (5h - SM) • 5. Rotation Dynamics (6h - SM) • 6. Mechanical Oscillators (8-11h) • 7. Energy Aspects (4-5h) • 8. Atom & Mechanics (5h - SM)

Unit 1: Newton's Laws		6 Hours	
Prerequisites: Vectors, velocity, acceleration, inertia principle, Galileo frames.		Problem Situation: Isaac Newton linked forces acting on a body to the motion of its center of mass. What are Newton's three laws and how do they predict motion on inclined planes?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Acceleration vector $a = dv/dt = d^2r/dt^2$. - Frenet frame components: $a = a_t \cdot u_t + a_n \cdot u_n = (dv/dt) \cdot u_t + (v^2/\rho) \cdot u_n$. - Newton's 1st Law (Inertia), 2nd Law ($\Sigma F_{ext} = m \cdot a_G$), 3rd Law (Action-Reaction). - Galilean reference frames and validity conditions. - Study motion of a solid on horizontal and inclined planes.	I. Kinematics Vectors (1. Position & velocity, 2. Acceleration vector, 3. Frenet frame) II. Newton's Laws (1. Newton's 1st Law, 2. Newton's 2nd Law $\Sigma F = m \cdot a_G$, 3. Newton's 3rd Law) III. Applications (1. Horizontal plane, 2. Inclined plane motion)	- Linear air track with optical photogates - Glider, calibrated traction masses, friction pads - Computer with Avimeca motion analysis software	Activities: Verify $a_G = F/m$ on air track. Project Newton's 2nd law along axes to compute friction coefficient. Evaluation: Inclined plane acceleration problems and Exam 4.

Unit 2: Vertical Fall of a Solid Body		4 Hours	
Prerequisites: Gravity field, Newton's 2nd law, integration.		Problem Situation: A parachutist accelerates before reaching a constant terminal velocity. What distinguishes free fall from fall in a fluid, and how does Euler's method solve it?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Free fall: only force is weight ($a = g$, $v(t) = g \cdot t$, $z(t) = 0.5 \cdot g \cdot t^2$). - Fall in fluid: weight P, Archimedes thrust F_A, and fluid friction $f = -k \cdot v^n$. - Differential equation: $dv/dt = A - B \cdot v$ (or v^2). - Terminal velocity: $v_{lim} = A / B$ when $dv/dt = 0$. - Euler's numerical method: $v_{i+1} = v_i + a_i \cdot \Delta t$ and $a_i = A - B \cdot v_i$.	I. Free Fall (1. Definition, 2. Differential equation $a = g$, 3. Time equations) II. Fall with Fluid Friction (1. Forces acting: P , F_A , f , 2. Differential equation $dv/dt = A - B \cdot v$, 3. Terminal speed v_{lim} , 4. Characteristic time τ) III. Euler's Numerical Method	- Tall glass cylinder filled with pure glycerol - Steel bearing balls of varied diameters - Digital camera, lighting screen, Avimeca and Regressi	Activities: Film ball drop in glycerol; plot $v(t)$; observe asymptotic approach to v_{lim} . Compute 5 steps of Euler table and compare with experiment. Evaluation: Euler table iterations and terminal velocity problems.

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Unit 3: Planar Motions - Projectiles and Charged Particles		5 Hours	
Prerequisites: Newton's 2nd law, vectors, uniform electric/magnetic fields.		Problem Situation: In soccer penalty kicks, initial velocity and launch angle decide whether a goal is scored. How does initial velocity govern parabolic trajectory, peak, and range?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Projectile parametric equations: $x(t) = (v_0 \cdot \cos \alpha) \cdot t$, $y(t) = -0.5g \cdot t^2 + (v_0 \cdot \sin \alpha) \cdot t$. - Trajectory parabola: $y(x) = -(g / (2v_0^2 \cdot \cos^2 \alpha)) \cdot x^2 + \tan(\alpha) \cdot x$. - Calculate trajectory peak (flèche) and impact range (portée). - Charged particle in magnetic field: Lorentz force $F = q(v \times B)$; circular uniform motion of radius $R = m \cdot v / (q \cdot B)$. - Charged particle in uniform electric field: parabolic deflection.	I. Projectile Motion (1. Newton's 2nd law application, 2. Parametric equations, 3. Cartesian trajectory, 4. Range & peak) II. Charged Particle in Uniform B (1. Lorentz magnetic force, 2. Proof of uniform circular motion, 3. Magnetic deflection) III. Charged Particle in Uniform E	- Spring cannon projectile launcher with angle scale - Fine electron beam deflection tube with Helmholtz coils - Video motion tracking software (Avimeca / Regressi)	Activities: Shoot projectile at 30°, 45°, 60°; measure range; verify 45° gives maximum range. Observe circular glow of electron beam in B field. Evaluation: Trajectory calculations, electron spectrometer problems, Exam 4.

Unit 4: Satellites and Planets (Mathematical Sciences)		5 Hours	
Prerequisites: Universal gravitation, circular motion, Frenet frame.		Problem Situation: The Moon orbits Earth every 27.3 days, while telecom satellites appear stationary. What are Kepler's three laws and what ensures geostationary orbit?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- State Kepler's 3 laws (Ellipses, Equal Areas, Periods: $T^2 / a^3 = \text{const}$). - Study circular orbital motion in Frenet frame: $a_n = v^2 / r$. - Orbital velocity: $v = \sqrt{G \cdot M / r}$. - Orbital period: $T = 2\pi \cdot \sqrt{r^3 / (G \cdot M)} \Rightarrow T^2 / r^3 = 4\pi^2 / (G \cdot M)$. - Geostationary satellite conditions (equatorial, West-to-East, $T = 24h$, $h \approx 36,000 \text{ km}$).	I. Kepler's Laws (1. Heliocentric frame, 2. 1st Law (Elliptical orbits), 3. 2nd Law (Law of areas), 4. 3rd Law (Law of periods)) II. Circular Orbital Motion (1. Newton's 2nd law, 2. Orbital velocity v , 3. Period T) III. Artificial Satellites (1. Geostationary satellite)	- Astronomy planetarium simulation software (Stellarium) - Planetary data tables (mass, radius, orbital semi-major axis, period)	Activities: Plot T^2 vs r^3 for solar system planets; verify that slope equals $4\pi^2 / (G \cdot M_{\text{Sun}})$. Calculate exact altitude of geostationary orbit. Evaluation: Orbital mechanics problem sets and Exam 5.

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Unit 5: Fundamental Dynamic Relation for Rotation (Mathematical Sciences)		6 Hours	
Prerequisites: Angular acceleration, moment of force, moment of inertia.		Problem Situation: Waterwheels accelerate when water torque exceeds resistive forces. What quantitative law links net torque to angular acceleration θ'' ?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Angular acceleration: $\theta'' = d\omega/dt = d^2\theta/dt^2$ (rad/s²). - Relation with linear acceleration: $a_t = R \cdot \theta''$ and $a_n = R \cdot \theta'^2$. - Fundamental Relation for Rotation: $\Sigma M_{\Delta}(F_{ext}) = J_{\Delta} \cdot \theta''$. - Moments of inertia for standard solids (disc, rod, cylinder, sphere). - Study mechanical systems with mixed translation and rotation.	I. Angular Kinematics (1. Angular velocity and acceleration, 2. Uniformly accelerated rotation: $\theta(t) = 0.5 \cdot \theta'' \cdot t^2 + \omega_0 \cdot t + \theta_0$) II. Fundamental Dynamic Law (1. Law statement $\Sigma M = J_{\Delta} \cdot \theta''$, 2. Moment of inertia J_{Δ} , 3. Applications)	- Rotational dynamics bench with precision optical pulley - Rotating disc, cylindrical ring, hanging acceleration mass - Electronic chronometer with infrared sensors	Activities: Measure angular acceleration θ'' for varied falling masses; verify that θ'' is proportional to net torque and inversely proportional to J_{Δ} . Evaluation: Dynamic rotation coupled problems and Exam 5.

Unit 6: Oscillating Mechanical Systems		8 Hours	
Prerequisites: Newton's 2nd law, rotation dynamics, Hooke's law, torque.		Problem Situation: In 1657, Huygens built the first pendulum clock. What characterizes mechanical harmonic oscillators (spring-mass, torsion, simple, physical)?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Elastic spring-mass oscillator: $m \cdot x'' + k \cdot x = 0$; period $T_0 = 2\pi \cdot \sqrt{m/k}$. - Torsion pendulum: $J_{\Delta} \cdot \theta'' + C \cdot \theta = 0$; period $T_0 = 2\pi \cdot \sqrt{J_{\Delta}/C}$. - Simple gravity pendulum: $\theta'' + (g/L) \cdot \sin(\theta) = 0$; small angles $T_0 = 2\pi \cdot \sqrt{L/g}$. - Physical pendulum: $J_{\Delta} \cdot \theta'' + m \cdot g \cdot d \cdot \theta = 0$; $T_0 = 2\pi \cdot \sqrt{J_{\Delta}/(m \cdot g \cdot d)}$. - Damping regimes and mechanical resonance phenomenon.	I. Spring-Mass Oscillator (1. Differential equation, 2. Harmonic solution $x(t)$, 3. Period $T_0 = 2\pi \sqrt{m/k}$) II. Torsion Pendulum (1. Equation $J_{\Delta} \cdot \theta'' + C \cdot \theta = 0$, 2. Period $T_0 = 2\pi \sqrt{J_{\Delta}/C}$) III. Gravity Pendulums (1. Simple pendulum $T_0 = 2\pi \sqrt{L/g}$, 2. Physical pendulum) IV. Damping & Resonance	- Horizontal air track with spring-mass glider - Torsion pendulum with metal bar and calibrated wire - Simple pendulum apparatus with angle protractor, photogate timer	Activities: Measure period T_0 of spring-mass vs mass m ; verify T_0^2 is proportional to m . Measure simple pendulum period vs length L to deduce g . Evaluation: Oscillator differential equations and period calculations.

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Unit 7: Energy Aspects of Mechanical Oscillators		4 Hours	
Prerequisites: Kinetic energy, potential energy, mechanical energy.		Problem Situation: Trapeze acrobats oscillate with continuous energy exchanges. What are the expressions for elastic potential energy and total mechanical energy?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Elastic potential energy of spring: $E_{pe} = 0.5 \cdot k \cdot x^2 + C$ (Joules). - Torsion potential energy: $E_{pt} = 0.5 \cdot C \cdot \theta^2 + C$ (Joules). - Mechanical energy: $E_m = E_c + E_p = 0.5 \cdot m \cdot v^2 + 0.5 \cdot k \cdot x^2 = \text{const.}$ - Derive differential equation by time differentiation of E_m: $dE_m/dt = 0$. - Energy diagrams $E_c(x)$, $E_p(x)$, $E_m(x)$ and potential wells.	I. Spring-Mass Energy (1. Work of spring tension, 2. Elastic potential energy E_{pe} , 3. Conservation of E_m) II. Torsion & Pendulum Energy (1. Torsion potential energy E_{pt} , 2. Gravitational potential energy E_{pp}) III. Energy Diagrams	- Horizontal spring glider with position and force sensors - Computer data acquisition system (Regressi/Datastudio)	Activities: Plot experimental curves $E_c(t)$, $E_p(t)$, and $E_m(t)$; verify total energy conservation; observe damped envelope with friction. Evaluation: Energy diagram reading problems and Supervised Exam 6.

Unit 8: The Atom and Newtonian Mechanics (Mathematical Sciences)		5 Hours	
Prerequisites: Planetary motion, Coulomb's law, light wave frequency, photon energy.		Problem Situation: Classical mechanics predicts orbiting electrons would radiate energy and collapse into the nucleus. How did Bohr's postulates introduce energy quantization?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Limits of Newtonian mechanics at microscopic atomic scale. - Concept of light photon: $E_{\text{photon}} = h \cdot \nu = h \cdot c / \lambda$ (Planck's constant h). - Quantization of atomic energy levels: $E_n = -E_0 / n^2$ (for hydrogen, $E_0 = 13.6$ eV). - Bohr's frequency condition: $\Delta E = E_p - E_n = h \cdot \nu$. - Hydrogen emission and absorption line spectra (Lyman, Balmer, Paschen).	I. Limits of Classical Mechanics (1. Rutherford planetary atom instability, 2. Continuous vs discrete energy) II. Energy Quantization (1. The photon, 2. Bohr postulates, 3. Energy levels of hydrogen) III. Atomic Line Spectra (1. Emission & absorption, 2. Balmer spectral series)	- Hydrogen discharge spectral tube, high-voltage power supply - Diffraction grating spectrometer (goniometer) - Solar Fraunhofer lines poster and energy diagram software	Activities: Observe hydrogen spectral lines through diffraction grating; calculate wavelengths of Balmer lines; match with ΔE transitions. Evaluation: Photon wavelength calculations and transition energy problem sets.

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Introduction & Part 1: Fast & Slow Reactions • Reaction Kinetics (Total: 8 - 11 Hours)

Questions Posed to the Chemist (1h): Understand chemical control, synthesis, tracking reaction evolution, and hazardous material safety.

Part 1 Units: 1. Fast and Slow Transformations (3h) • 2. Time Tracking of a Chemical Reaction - Reaction Rate (5-8h)

Unit 1: Fast and Slow Transformations 3 Hours

Prerequisites:

Oxidation-reduction, redox couples Ox/Red, electron transfer.

Problem Situation:

Book pages slowly yellow over decades, while gasoline combustion occurs in milliseconds. What separates slow and fast reactions, and what are kinetic factors?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Fast reactions: instantaneous change impossible to track with human eye. - Slow reactions: progress measurable over seconds, minutes, or hours. - Kinetic factors: temperature (thermal quenching / acceleration). - Initial reactant concentrations effect. - Role of catalysts in speeding reactions without consumption.	I. Redox Recall (1. Couples Ox/Red, 2. Half-equations) II. Fast vs Slow Reactions (1. Fast precipitation, 2. Slow oxidation of I^- by H_2O_2) III. Kinetic Factors (1. Temperature effect, 2. Concentration effect, 3. Catalysts)	- Solutions: KI, H_2O_2, $Na_2S_2O_3$, starch indicator, ice bath - Test tubes, beakers, magnetic stirrer, hot water bath	Activities: Mix I^- and H_2O_2 at $0^\circ C$, $20^\circ C$, $50^\circ C$; observe blue starch appearance time. Demonstrate quenching in ice water. Evaluation: Redox equations balancing and kinetic factors tests.

Unit 2: Time Tracking of a Chemical Reaction - Reaction Rate 6 Hours

Prerequisites:

ICE table, conductometry, ideal gas law, titration.

Problem Situation:

Leaves gradually lose their green color in autumn. How do chemists track the exact progress $x(t)$ over time, and what is the half-life $t_{1/2}$?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Methods of kinetic tracking: chemical titration, conductometry, manometry, spectrophotometry. - Volumetric reaction rate: $v(t) = (1/V) \cdot (dx/dt)$ ($mol \cdot L^{-1} \cdot s^{-1}$). - Determine reaction rate graphically from slope of tangent to $x(t)$. - Rate decreases over time due to depletion of reactants. - Half-life of reaction $t_{1/2}$: $x(t_{1/2}) = x_f / 2$.	I. Tracking Methods (1. Chemical titration with quenching, 2. Physical: conductometry & pressure) II. Volumetric Reaction Rate (1. Definition $v = (1/V) dx/dt$, 2. Tangent method, 3. Rate decrease interpretation) III. Reaction Half-Life (1. Definition $x(t_{1/2}) = x_f / 2$, 2. Graphical extraction)	- Precision conductometer, sealed flask with manometer - Burettes, pipettes, stopwatch, ice bath - Reagents: HCl, Mg ribbon, KI, H_2O_2, $Na_2S_2O_3$	Activities: Track $Mg + 2HCl$ by measuring H_2 pressure $P(t)$; express $x(t) = \Delta P \cdot V / (R \cdot T)$; plot $x(t)$ and measure tangents to calculate $v(0)$ and $v(t)$. Find $t_{1/2}$. Evaluation: Tangent slope calculations and half-life determinations; Exam 1.

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Part 2: Non-Total Transformations of a Chemical System (Total: 13 - 17 Hours)

Unit 1: Chemical Reactions Occurring in Both Directions		3 Hours	
Prerequisites: ICE table, limiting reactant, Brønsted acid-base theory.		Problem Situation: Cave formation dissolves limestone, while cave stalactite precipitation reverses it. What characterizes reactions that occur in both directions?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Measure solution pH: $\text{pH} = -\log[\text{H}_3\text{O}^+] \Leftrightarrow [\text{H}_3\text{O}^+] = 10^{(-\text{pH})}$. - Final progress x_f vs theoretical maximum progress x_{max}. - Final progress ratio: $\tau = x_f / x_{\text{max}}$. - Distinguish total reaction ($\tau = 1$) from limited reaction ($\tau < 1$). - State of dynamic chemical equilibrium (two opposing directions at equal rates).	I. Acid-Base & pH (1. Brønsted couples, 2. Definition & measurement of pH) II. Progress of Reaction (1. Maximum progress x_{max} , 2. Final progress x_f , 3. Final progress ratio τ) III. State of Chemical Equilibrium (1. Equilibrium state, 2. Microscopic interpretation)	- Calibrated digital pH-meter with buffer solutions (pH 4, 7) - Solutions: Ethanoic acid CH_3COOH, hydrochloric acid HCl (same C) - Magnetic stirrer, beakers, precision volumetric pipettes	Activities: Measure pH of 0.01 M HCl (find $\text{pH} \approx 2$, $\tau = 1$, total) vs 0.01 M CH_3COOH (find $\text{pH} \approx 3.4$, $\tau \approx 4\%$, limited equilibrium). Evaluation: Calculations of x_f , x_{max} , and τ from pH.

Unit 2: State of Chemical Equilibrium		5 Hours	
Prerequisites: Final progress ratio τ , conductivity of ionic solutions.		Problem Situation: Salicylic acid in willow bark is in chemical equilibrium with water. What is the equilibrium constant K and what factors influence final progress?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Reaction quotient Q_r expression: $Q_r = \Pi [\text{Products}]^p / \Pi [\text{Reactants}]^r$. - Solids and solvent water take activity equal to 1 in Q_r. - Equilibrium constant: $K = Q_{r,\text{eq}}$ depends exclusively on temperature. - Independence of K from initial concentrations. - Influence of initial concentration: dilution increases final progress ratio τ.	I. Reaction Quotient Q_r (1. Definition, 2. Homogeneous vs heterogeneous systems, 3. $Q_{r,\text{eq}}$ at equilibrium) II. Equilibrium Constant K (1. $K = Q_{r,\text{eq}}$ definition, 2. Total reaction $K > 10^4$, 3. Reverse reaction constant $K' = 1/K$) III. Factors Influencing τ (1. Initial state effect, 2. Equilibrium constant effect)	- Precision conductometer with dip cell, digital thermometer - Solutions: Ethanoic acid CH_3COOH at 10^{-1} M, 10^{-2} M, 10^{-3} M - Methanoic acid HCOOH solutions, magnetic stirrer	Activities: Measure conductivity σ for 3 concentrations of ethanoic acid; calculate $Q_{r,\text{eq}}$; verify that K is identical while τ increases with dilution. Evaluation: Equilibrium constant K calculations and Supervised Exam 2.

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Part 2: Chemistry • Unit 3: Acid-Base Reactions in Aqueous Solution (8 Hours)

Unit 3: Acid-Base Transformations in Aqueous Solution		8 Hours	
Prerequisites: pH, equilibrium constant K, Brønsted acid-base theory.		Problem Situation: Pharmaceutical research requires precise pH-metric titrations of acidic drugs. What is the acidity constant K_a , and how is the equivalence point located?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Water auto-ionization: $2 \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{HO}^-$; ionic product $K_e = [\text{H}_3\text{O}^+][\text{HO}^-] = 10^{-14}$ at 25°C. - Acidity constant of couple: $K_a = [\text{A}^-][\text{H}_3\text{O}^+] / [\text{HA}]$ and $\text{p}K_a = -\log(K_a)$. - Fundamental Henderson relation: $\text{pH} = \text{p}K_a + \log([\text{A}^-] / [\text{HA}])$. - Predominance diagrams: $[\text{HA}] > [\text{A}^-]$ if $\text{pH} < \text{p}K_a$; $[\text{A}^-] > [\text{HA}]$ if $\text{pH} > \text{p}K_a$. - Acid-base titration curve $\text{pH} = f(V_b)$: equivalence point E by parallel tangent method. - Select appropriate colored indicator: indicator $\text{p}K_{\text{In}}$ zone covers pH_E.	I. Ionic Product of Water (1. Auto-ionization K_e, 2. Neutral, acidic, basic solutions) II. Acidity Constant K_a & $\text{p}K_a$ (1. Definition, 2. Relation $\text{pH} = \text{p}K_a + \log([\text{A}^-]/[\text{HA}])$, 3. Acid-base reaction constant $K = K_{a1} / K_{a2}$) III. Predominance Diagrams (1. Couples HA/A^-, 2. Colored indicators) IV. pH-Metric Titrations (1. Titration curves, 2. Equivalence point E, 3. Tangent method & dpH/dV derivative, 4. Indicator selection)	- Digital pH-meter with glass electrode and buffers (4.00, 7.00) - Precision burette (25 mL), volumetric pipette (10 mL), magnetic stirrer - Solutions: Ethanoic acid, hydrochloric acid, sodium hydroxide NaOH - Indicators: Bromothymol blue, phenolphthalein, methyl orange	Activities: Perform pH-metric titration of CH_3COOH with NaOH; record pH after each 0.5 mL; plot $\text{pH} = f(V_b)$; locate E by tangent method and compare with derivative dpH/dV peak. Evaluation: pH-metric titration curves and indicator justification problems; Exam 2/3.

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Part 3: Direction of Evolution of a Chemical System (Total: 8 - 18 Hours)

Unit 1: Spontaneous Evolution of a Chemical System			3 Hours
Prerequisites: Reaction quotient Q_r , equilibrium constant K .		Problem Situation: Coral skeletons form from calcium and carbonate ions, but can dissolve when ocean concentrations shift. How do chemists predict the direction in which a system evolves?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Calculate initial reaction quotient $Q_{r,i}$ from initial concentrations. - Spontaneous Evolution Criterion: compare $Q_{r,i}$ with K. - If $Q_{r,i} < K$: evolves spontaneously in forward direction (1). - If $Q_{r,i} > K$: evolves spontaneously in reverse direction (2). - If $Q_{r,i} = K$: system is at equilibrium (no macroscopic evolution).	I. Recall of Q_r & K (1. Initial quotient $Q_{r,i}$, 2. Equilibrium constant K) II. Spontaneous Evolution Criterion (1. Forward direction ($Q_{r,i} < K$), 2. Reverse direction ($Q_{r,i} > K$)) III. Applications (1. Acid-base mixtures, 2. Redox mixtures (Fe^{2+}/Fe^{3+} , I_2/I^-))	- Solutions: I_2, KI, $FeSO_4$, $Fe_2(SO_4)_3$, $AgNO_3$ - Beakers, test tubes, copper wire, spectrophotometer	Activities: Mix solutions with $Q_{r,i} > K$; verify formation of reactants and disappearance of products. Mix with $Q_{r,i} < K$ to confirm forward shift. Evaluation: Direction prediction calculations and Exam 3.

Unit 2: Spontaneous Reactions in Voltaic Cells and Energy Recovery			5 - 7 Hours
Prerequisites: Redox reactions, electric current, Faraday's constant.		Problem Situation: Batteries power cars and portable electronics. How do spontaneous chemical reactions in electrochemical cells generate electric power?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Direct electron transfer (in same beaker) vs indirect transfer (via wire). - Constitution of galvanic cell: 2 half-cells, electrodes, and salt bridge. - Anode (-): Oxidation (loss of electrons). Cathode (+): Reduction. - Cell electrical capacity: $Q_{max} = I \cdot \Delta t = n(e^-) \cdot F$. - Faraday constant: $F = N_A \cdot e \approx 96500 \text{ C/mol}$. - Relate mass changes Δm of electrodes to current I and duration Δt.	I. Electron Transfer (1. Direct transfer, 2. Indirect transfer through circuit) II. Galvanic Cell (Daniell Cell) (1. Structure & salt bridge role, 2. Anode & Cathode half-reactions, 3. Standard cell diagram) III. Electrical Capacity & Matter Balance (1. $Q = I \cdot \Delta t = n(e^-) \cdot F$, 2. Electrode mass variation)	- Copper plate, zinc plate, salt bridge (agar-agar + KNO_3) - Solutions: $CuSO_4$ (1 mol/L) and $ZnSO_4$ (1 mol/L) - Milliammeter, digital voltmeter, resistor (100 Ω), leads	Activities: Build Daniell cell; measure open-circuit EMF ($E \approx 1.1 \text{ V}$); observe electron flow from Zn to Cu; calculate mass of consumed Zn. Evaluation: Cell equations, electrode polarity, and battery capacity calculations.

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Part 3: Chemistry • Unit 3: Forced Transformations - Electrolysis (8 Hours)

Unit 3: Forced Transformations - Electrolysis			8 Hours
Prerequisites: Redox reactions, electric current, electrochemical cells.		Problem Situation: Pure copper is extracted and recycled from ores by electrolysis. How does an external generator force a non-spontaneous chemical reaction to occur?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Definition of electrolysis: forced endergonic chemical transformation. - External DC generator imposes electron flow opposite to spontaneous direction. - Anode (+): Oxidation (connected to positive terminal). - Cathode (-): Reduction (connected to negative terminal). - Quantitative law: $m = (I \cdot \Delta t \cdot M) / (n \cdot F)$. - Industrial applications: electroplating, copper refining, chlorine production.	I. Principle of Electrolysis (1. Spontaneous vs forced transformations, 2. Electrolyzer circuit & DC generator, 3. Electrode reactions) II. Quantitative Study (1. Charge $Q = I \cdot \Delta t = n(e^-) \cdot F$, 2. Mass of deposited metal, 3. Volume of evolved gas) III. Industrial Applications (1. Lead accumulator recharging, 2. Electroplating)	- U-shaped electrolysis tube, graphite electrodes, copper electrodes - Copper(II) chloride solution CuCl_2, sodium chloride solution - Adjustable DC power supply (0-12V, 2A), ammeter, stopwatch	Activities: Electrolyze CuCl_2 solution; observe chlorine gas bubbling at anode and copper metal deposition at cathode. Weigh cathode before and after to verify Faraday's mass law. Evaluation: Faraday electrolysis calculations and Supervised Exam 3.

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Part 4: Controlling the Evolution of Chemical Systems (Total: 9 - 12 Hours)

Unit 1: Esterification and Hydrolysis Reactions			5 Hours
Prerequisites: Alcohols, carboxylic acids, functional groups, equilibrium.		Problem Situation: Beehive wax and honey aromas are formed from esters. How do carboxylic acids and alcohols react to produce esters, and how can the limited yield be improved?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Identify ester group -COO- and systematic IUPAC nomenclature. - Esterification equation: $R-COOH + R'-OH \rightleftharpoons R-COO-R' + H_2O$. - Characteristics: slow, reversible, athermic, limited yield. - Hydrolysis of ester: reverse limited reaction. - Reaction yield: $r = n_{\text{exp}}(\text{ester}) / n_{\text{max}} = x_f / x_{\text{max}}$. - Improve yield: excess reactant, Dean-Stark water removal, or using acid anhydride.	I. Oxygenated Organic Families (1. Alcohols, 2. Carboxylic acids, 3. Acid anhydrides, 4. Esters & nomenclature) II. Esterification & Hydrolysis (1. Experimental study, 2. Kinetics & equilibrium constant $K \approx 4$ (for 1° alcohol), 3. Yield r) III. Controlling Reaction (1. Catalyst H_2SO_4 & temperature effect, 2. Shifting equilibrium)	- Reflux heating setups with Liebig condensers, heating mantles - Ethanol, ethanoic acid, sulfuric acid H_2SO_4, ice bath - Burettes, NaOH titrant solution, phenolphthalein	Activities: Synthesize ethyl ethanoate under reflux; titrate remaining acid over time; plot esterification progress curve; calculate yield r ($\approx 67\%$ for equimolar primary alcohol). Evaluation: Ester nomenclature, yield optimization calculations, Exam 4.

Unit 2: Controlling Chemical Reactions - Saponification			4 Hours
Prerequisites: Esters, acid anhydrides, basic solutions, reaction yield.		Problem Situation: Soap is manufactured by alkaline hydrolysis of fatty triglycerides. How is soap prepared with 100% yield, and what makes soap clean grease?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Fast and total ester synthesis using acid anhydride: $(R-CO)_2O + R'-OH \rightarrow R-COOR' + R-COOH$. - Basic hydrolysis of esters (saponification): $R-COOR' + HO^- \rightarrow R-COO^- + R'-OH$ (total reaction). - Soap synthesis from fatty triglycerides (vegetable oils/fats) + 3 NaOH $\rightarrow$ glycerol + 3 soap molecules. - Salting-out process in saturated NaCl brine. - Amphiphilic nature of soap carboxylate ions: hydrophobic hydrocarbon tail and hydrophilic polar head.	I. Synthesis Using Anhydrides (1. Acid anhydride reactivity, 2. Total synthesis) II. Basic Hydrolysis (Saponification) (1. Saponification equation, 2. Triglycerides & fats, 3. Total & fast characteristics) III. Soap Properties (1. Salting-out in brine, 2. Hydrophilic / hydrophobic structure, 3. Micelle mechanism)	- Olive oil, commercial ethanol, 10 M sodium hydroxide NaOH - Saturated salt brine (NaCl solution), Büchner filter apparatus - Heating mantle, round-bottom flask, boiling stones, watch glasses	Activities: Heat olive oil with NaOH in ethanol under reflux for 30 min. Pour into cold saturated brine to precipitate soap (salting out). Filter on Büchner funnel and test lathering with water. Evaluation: Saponification mass yield calculations and end-of-year Baccalaureate Exam.